EXPERIMENT TO UNDERSTAND K-MEANS CLUSTRING ALGORITHMN FOR A GIVEN DATASET

PROGRAM:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

%matplotlib inline

df = pd.read_csv('Mall_Customers_modified.csv')
df.info()
df.head()
sns.pairplot(df)
features = df.iloc[:, [3, 4]].values
Features
from sklearn.cluster import KMeans

model = KMeans(n_clusters=5)
model.fit(features)
Final = df.iloc[:, [3, 4]].copy()
Final['label'] = model.predict(features)
Final.head()
sns.set_style("whitegrid")
```

```
sns.FacetGrid(Final, hue="label", height=8) \
.map(plt.scatter, "Annual Income (k$)", "Spending Score (1-100)") \
.add_legend()

plt.show()
features_el = df.iloc[:, [2, 3, 4]].values

from sklearn.cluster import KMeans

wcss = []

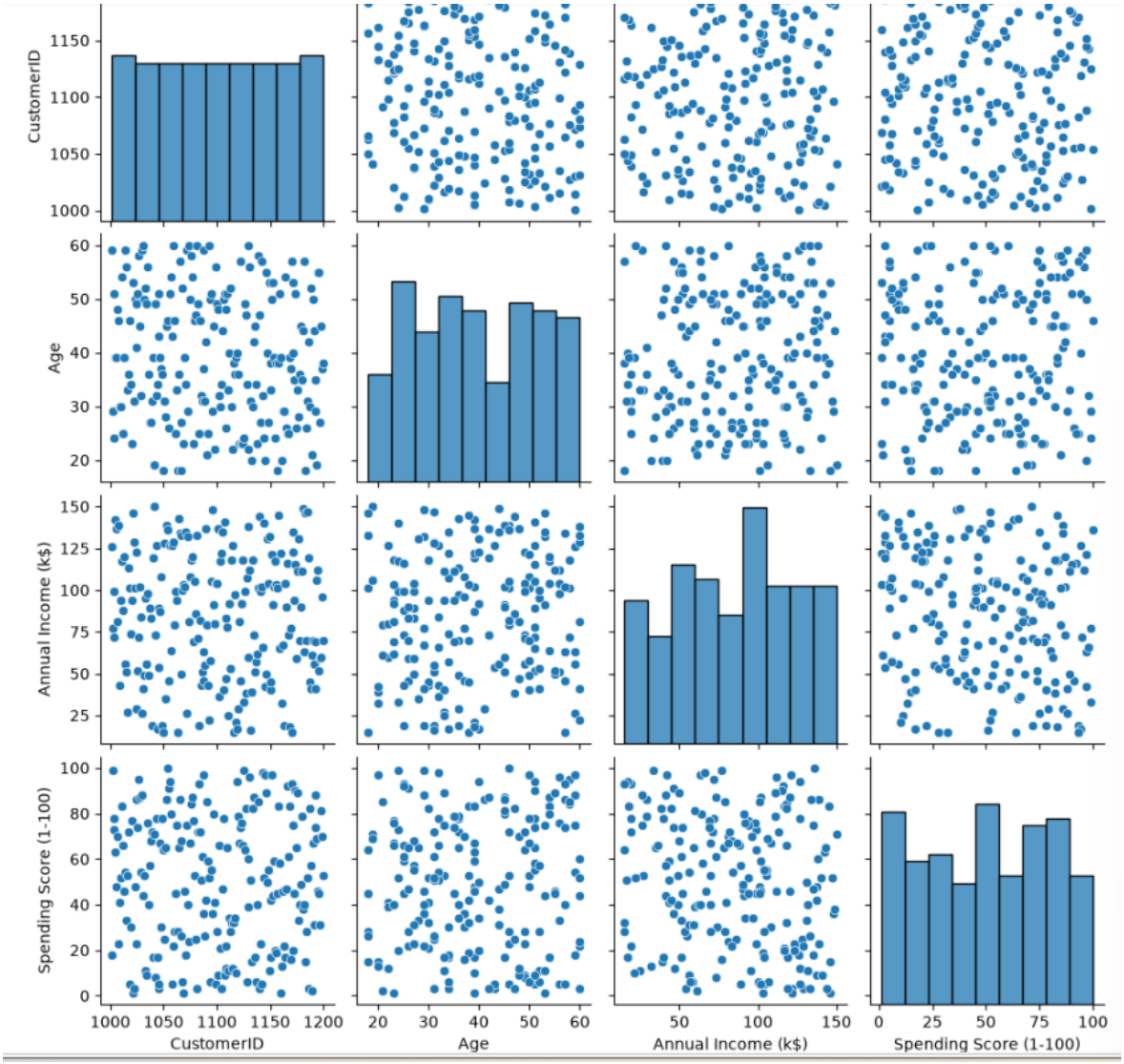
for i in range(1, 10):
    model = KMeans(n_clusters=i)
    model.fit(features_el)
    wcss.append(model.inertia_)
plt.plot(range(1, 10), wcss)
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.show()
```

OUTPUT:

```
<class 'pandas.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   CustomerID              200 non-null    int64
 1   Gender                  200 non-null    str
 2   Age                     200 non-null    int64
 3   Annual Income (k$)      200 non-null    int64
 4   Spending Score (1-100)  200 non-null    int64
dtypes: int64(4), str(1)
```

```
memory usage: 7.9 KB
```

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1001	Female	59	126	18
1	1002	Male	29	77	99
2	1003	Female	24	99	73
3	1004	Male	51	72	78
4	1005	Female	39	142	63



```
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```

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       [ 70,  53]])
```

	Annual Income (k$)	Spending Score (1-100)	label	
0	126	18	1	
1	77	99	2	
2	99	73	2	
3	72	78	2	
4	142	63	3	

